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The Effect of Property Taxes on Home Values

Jeffrey P. Guilfoyle*

Abstract

This article examines the theoretical and empirical economic literature of the effects of property taxes on the price of residential housing. The results in the theoretical literature are largely driven by the various modeling assumptions. Therefore, the effect must be measured empirically. Unfortunately, the empirical work has yielded contradictory results. It is unclear if the empirical differences in recent studies are the result of methodological differences or if other factors are at work. The empirical work needs to incorporate additional variables that may be influencing the capitalization effect in an attempt to identify the other factors that may be at work.

Introduction

In this article, the economic literature of the effects of the property tax on the price of residential housing is reviewed. The article is divided into two main sections. First, the theoretical literature on the impact of property taxes on house prices is examined. Second, the empirical literature on the impact of property taxes on house values is examined.

If a property tax differential is reflected in the prices of otherwise identical properties, the tax differential is said to be capitalized. The degree to which property taxes are capitalized is an important element in examining property tax policies for several reasons. First, the degree to which the tax is capitalized has an effect on the incidence of the property tax. If a property tax change is fully capitalized, the selling value of the asset is reduced by the present discounted value of the tax. Under full capitalization, the owners of a property at the time of a tax change bear the full burden of the tax. They are unable to escape the tax by selling their property because the property's value has been reduced by the full amount of the future tax stream.

Second, capitalization is important for determining the efficiency of the property tax. The degree to which the property tax is a distorting tax on capital is in part determined by the degree of capitalization. The ability of local governments to provide local public goods efficiently, and the amount of redistribution that occurs through a local public good provision, are also in part determined by the degree of property tax capitalization. Finally, different economic theories of property taxes have somewhat different predictions concerning the extent to which property taxes are capitalized. Examining the degree to which a property tax change is capitalized can help in the evaluation of these theories.

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The Theory of the Property Tax

The Benefit View

Although a decentralized market system has many advantages, it generally will not produce the optimal level of public goods. In his classic article, Samuelson (1954: 388) argues that "no decentralized pricing system can serve to determine optimally these levels of collective consumption." Writing in response to Samuelson, Tiebout (1956) argues that while this may be true for public goods provided at the federal level, it is not necessarily true for local public goods. Tiebout develops a list of assumptions under which it was possible for local public goods to be provided at an efficient level. Tiebout's model consists of a large number of local governments that finance expenditures via a head tax. Consumers reveal their preferences for local goods through their choice of residence. Consumers preferring a high level of public services live in a locality with high taxes and services. Consumers preferring a low level of services live in a locality with low taxes and services.

Tiebout's model contains a number of highly restrictive assumptions. One of these assumptions is that local expenditures can be financed via a head tax.[1] Subsequent authors have examined the implications of replacing Tiebout's head tax with the more commonly observed property tax. This modeling change has led to the "benefit" view of the property tax. In this section, the benefit view of the property tax is discussed. This view argues that the property tax does not necessarily lead to inefficiencies.

Hamilton (1976) argues for the benefit view of property taxation. He first notes that, in a system of communities that are homogeneous with respect to house values, the property tax can be seen as a system of average cost pricing for public services. Hamilton assumes that all households in a community consume the same level of public services and place the same value on these services. Homogeneous communities can be achieved through a binding zoning requirement. That is, all households in a community are required to consume a minimum amount of housing services, the tax on which is exactly equal to the cost of providing public services to each household. No household has any incentive to consume more than the minimum level of housing services because they would then be paying for more services than they receive (and would be better off in another community). If the zoning requirement is in fact binding and communities are homogeneous, public services are provided efficiently and the property tax is non-distortionary. The zoning requirement keeps households from adjusting their housing consumption in response to the tax.

The binding zoning requirement has been strongly criticized by Zodrow and Mieszkowski (1983). They argue that these assumptions would turn any tax into a non-distortionary tax. The choice of tax instrument at the local level would become irrelevant with the appropriate zoning restrictions since all taxes would become a non-distortionary fee for public service.

Hamilton, however, extends his argument to communities that are not homogeneous. He argues that capitalization effects can lead to average-cost pricing for public

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services in non-homogeneous communities. In a neighborhood with inexpensive and expensive houses and no capitalization, residents of the inexpensive houses enjoy public services at a lower cost than their neighbors in the expensive housing. This makes living in a relatively inexpensive house desirable. Therefore, people will compete to live in fiscally advantaged housing, driving up the price of such housing. Thus, inexpensive houses, although they have lower property taxes, sell for more due to the capitalization of the fiscal benefits. Likewise, housing that is relatively luxurious compared to other houses in the same community will sell at a discount.

If differences in fiscal surpluses are fully capitalized, then there is no advantage to buying an inexpensive house in a rich neighborhood. The price of the inexpensive house is increased by the present discounted value of the difference between its taxes and service cost. This has potentially important implications. First, it means that property taxes do not lead to horizontal inequity. Everyone gets exactly the services that they pay for. Second, because people pay for exactly the services that they receive, they demand an efficient level of public services.

For the property tax to be non-distortionary, however, it must also not distort housing decisions. Hamilton (1983) argues that the property tax can be converted into an efficient price for public services if capitalization causes the following relationship to hold.

$$V + T = C(H) + C(LPS). \tag{1}$$

Equation (1) states that efficiency requires that house value, V, plus taxes, T, must equal the cost of providing housing, $C(H)$, plus the cost of providing local public services, $C(LPS)$. If this relationship is to hold, then fiscally advantaged housing, which has a higher value, must somehow cost more to produce. In Hamilton's model (1976), fiscally advantaged housing costs more to produce because its fiscal advantage is capitalized into land values. Inexpensive housing is built on land zoned for that purpose. This land, because it can be used for houses that will sell at a premium, sells for a higher price than other land in the community. Hamilton implicitly assumes that property taxes are fully capitalized into land values. He assumes that land is in fixed supply; therefore, the tax is not distortionary.

This analysis would imply that land zoned for inexpensive housing in a heterogeneous community would sell for more than land zoned for more luxurious housing. Zoning ordinances are required to keep this land price differential in place. Otherwise, the amount of land used for inexpensive housing will increase and the land devoted to relatively expensive housing will decrease. If inexpensive housing is allowed to expand until the land for inexpensive housing and the land for relatively more expensive housing sells at the same price, households owning cheaper houses will pay less for their public services than households owning relatively expensive houses. Redistribution would occur through the provision of public services. Residents no longer get only what they pay for, and the outcome can no longer be expected to be efficient.

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Based on his analysis, Hamilton draws the following conclusions. First, with full capitalization, there is no horizontal inequity. Consumers get the services that they pay for. Second, efficient supplies of housing and public services exist when: land value is the same in all homogeneous (with respect to house value) communities; in mixed-value communities, land-value differentials exactly reflect the present value of fiscal surplus differentials; the mean value of land (per acre) is the same in all communities, regardless of their housing or service mix. Finally, the property tax generates an incentive for the production of an inefficient amount of low-income housing. This incentive must be held in check through zoning restrictions for efficiency to be maintained.

Hamilton cites some evidence that the above conditions may in fact hold. He argues (1976) that empirical studies have shown that land zoned for low-income housing does indeed sell at a premium. He also cites empirical studies (1983) that find a large degree of property tax capitalization.

The New and Classical Views

An alternative to the benefit view of property taxes is the "new" view. The new view of the property tax was developed by Thomson (1965), Mieszkowski (1972) and Aaron (1975). The discussion of the new view of the property tax presented here is taken primarily from a review of the property tax literature by Mieszkowski and Zodrow (1989) and a discussion of the new view in Zodrow and Mieszkowski (1983).

The new view of the property tax generally examines the property tax in a standard capital taxation framework. These models follow the Harberger (1962) approach to examining the taxation of capital. Harberger models generally assume perfect competition and that the overall capital stock in the nation is fixed. Additionally, the capital stock is assumed to be perfectly mobile within the nation, so that the after-tax return to capital in all sectors of the economy is the same.

Housing is viewed as the output of a production process that combines land and capital in the production of housing services. These models examine the effects of a property tax in one or more sectors. Although the results vary somewhat depending on the model specifications, they often produce the result that the capital portion of a uniform national property tax reduces the after-tax rate of return to *all* capital in the nation by the amount of the property tax. The property tax impacts other sectors in the economy because capital is perfectly mobile. When the property tax is imposed, it reduces the return to capital in the housing sector. Capital flows from the housing sector, increasing its pre-tax rate of return, into other sectors, reducing their rate of return. This continues until all sectors have the same after-tax return to capital. Capital generally bears the full brunt of taxation in these models because it is assumed that the total capital stock is supplied inelastically.

If the property tax were the only tax in the economy, it would have two effects. First, the property tax would result in an inefficiently low allocation of capital to the housing sector. Second, the overall rate of return to capital in the economy would be reduced.

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This rate reduction would have an effect on the long-run capital formation in the economy. This long-run effect is the subject of much debate in the literature and is beyond the scope of this analysis.

Of course, there are other taxes on capital, most notably the corporate income tax. Gravelle (1994) argues that housing, especially owner-occupied housing, is taxed at a very low rate compared to other forms of capital. She cites studies that find that the preferential treatment of housing accounts for about half of the distortions arising from the misallocation of capital. Because of this, the property tax may actually help to correct the misallocation of resources resulting from the corporate income tax.

The new view also has implications for the redistributional effects of the property tax. The property tax finances local public expenditures by reducing the rate of return of capital in all sectors of the economy. This means that, for a uniform national property tax, capital owners bear the full burden of the tax on the capital portion of housing. Because capital tends to be concentrated in the hands of relatively wealthy individuals, the capital portion of the property tax is progressive. This result directly conflicts with Hamilton's findings that no redistribution occurs through the property tax.

The property tax, of course, is not uniform throughout the nation. Under the new view, tax differentials between communities (that is, differences between local tax rates and the national average tax rate) give rise to "excise" effects. A tax rate higher than the average will be either shifted forward into higher housing prices, or shifted backward to a relatively immobile factor. Different models have different predictions as to the degree to which this will occur. The predictions of these models generally depend on the elasticity of substitution between land and capital in the production of housing, and assumptions concerning the mobility of consumers and the degree to which land is in fixed supply.

For example, consider property that rents for $100 per year. If the discount rate is 10% and we assume the asset is infinitely lived, the present discounted value of this property, in the absence of taxation, is $1000. If we impose a property tax of $10 per year and residents are completely immobile, we might see the tax fully forward shifted to renters. This means that the cost of renting this property would rise to $110 and the value of the property would remain unchanged. If renters are fully mobile, however, we might see the tax fully backward shifted into the property's price. Backward shifting means that the property still rents for $100, but now the owner must pay $10 in tax. The net return is $90 per year, reducing the present discounted value of the property to $900. In the case of full backward shifting of the tax, we say that the tax has been *fully capitalized.*

The predictions of the new view models are based on the assumptions made concerning various parameter values. For example, Hobson (1986) finds that the degree of property tax shifting depends in part on the relative sizes of the elasticity of substitution between housing and other consumption goods, and the elasticity of substitution in housing production, as well as the mobility of the resident population.[2]

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In their review of the property tax literature, Mieszkowski and Zodrow (1989) argue that the new view is a more general case of the "classical" view of the property tax. The classical view (Simon, 1943; and Netzer, 1966) examines the property tax in a single jurisdiction. This view argues that, since capital is perfectly mobile, it bears none of the burden of the property tax. The capital portion of the property tax is assumed to be entirely forward shifted in the form of higher housing prices, and the land portion is assumed to be borne entirely by landowners because land is assumed to be inelastic in supply. Under this view, the property tax is much more regressive than under the new view, because a much higher proportion is borne by renters.

Mieszkowski and Zodrow argue that the classical view focuses exclusively on the excise effects portion of the property tax. This arises because the classical view uses a partial rather than a general equilibrium model. The full forward shifting found in the classical view is just a special case of the possible results predicted by the new view. Although the overall rate of return on capital appears unchanged in the classical view, Mieszkowski and Zodrow argue that it is in fact reduced by an infinitesimal amount. This reduction occurs because the higher property tax rate in the metropolitan region reduces the national average by a small amount. This infinitesimal change multiplied by the entire capital stock of the nation is in fact large relative to the revenue raised by the tax.

Mieszkowski and Zodrow argue that, under the new view, there is a tendency towards under-provision of local public services. This occurs because local jurisdictions are reluctant to tax mobile capital. Additionally, local jurisdictions do not take into account the possible externalities associated with taxing mobile capital. Capital that flees a jurisdiction to escape a local property tax benefits other jurisdictions. This reduces the overall cost of taxing mobile capital in a way not taken into account by the taxing jurisdiction. Again, they find that these results are sensitive to model specification.

Yinger's Theory of Capitalization

Yinger (1982) develops a model in which he derives a household's bid for housing. He argues that the amount a household is willing to pay for a unit of housing services, in a particular jurisdiction, is based on the jurisdiction's level of services and taxes. Therefore, a household's bid for housing services can be written as $P = P(E, t)$, where E is the level of local public services per household and t is the effective property tax rate. The value of a house to a given household can be written as:

$$V(E, t) = P(E, t)H/r, \tag{2}$$

where H is the total amount of housing units consumed and r is the discount rate.[3]

Household utility is assumed to be a function of housing services, the level of local public services and a composite private good, Z. The household is assumed to maximize this utility function subject to its budget constraint. In deriving the first-order conditions, Yinger assumes that the household is choosing Z, H, E and t.

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The amount a household is willing to bid for housing, $P(E, t)$, is derived by solving for P from the first-order conditions of the utility maximization problem. Solving for $P(E, t)$ requires solving two first-order differential equations, and requires the assumption of a specific utility function. Additionally, the solution to $P(E, t)$ contains a constant of integration.

If a Cobb Douglas utility function of the form, $U = c_1\ln(Z) + c_2\ln(H) + c_3\ln(E)$, is assumed, and if housing services are assumed to be a multiplicative function of housing characteristics, X_1 to X_M, the value of housing can be expressed in the following form:[4]

$$\ln(V) = \ln(v) + (c_3/c_2)\ln(E) - \ln(r + t) + \Sigma a_i\ln(X_i). \tag{3}$$

Based on this derivation, Yinger draws a number of conclusions. First, differences in service levels between jurisdictions will be inexactly capitalized. The degree of capitalization will be based on taste parameters in the utility function. Second, because there are no taste parameters on the tax term in the value equation, differences in tax rates between communities will be exactly capitalized—regardless of the tastes of consumers.

Interestingly, Yinger also claims that his results imply that homogeneous communities can be formed without zoning barriers. If P_E, the derivative of the bid for housing with respect to services, increases with income, Yinger argues that high-income households will outbid low-income households for housing in high-service jurisdictions, so that zoning barriers are not required to sustain the mean property tax base.

This conclusion, however, does not follow from Yinger's model. If the amount of housing required to enter a community were fixed, as in a zoning requirement, then it is true that rich households would outbid poor households. Yinger's model does not impose this restriction, however. Households are free to choose the level of housing they consume. Rich and poor households would probably select very different levels of housing, given a tax and service package. The assertion that the rich will bid more for one unit of housing is not relevant since households will consume different amounts of housing. Yinger's model does not prevent the conclusion that poor families will try to consume a small amount of housing in a high-service jurisdiction.

Yinger recognizes that if the price for housing is different in different jurisdictions, suppliers will have an incentive to supply houses to jurisdictions with high house values. However, Yinger concludes: ". . . the supply of land within a jurisdiction is fixed, so the conversion of nonresidential land into residential land cannot continue indefinitely. Once all profitable conversion has occurred—that is, in long-run equilibrium—local fiscal variables will be capitalized into house values."

Developers will not simply continue to add new jurisdictions, because there are other factors affecting house values besides the service/tax package. Expanding further from

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the city center will cause the price households are willing to pay for housing services to fall, ceteris paribus. This is due to the increase in commuting costs for residents in more distantly located suburbs. Profits for housing developers will fall as they continue to develop further and further from the city center.[5]

Yinger specifies an equilibrium condition for determining the furthest point of community expansion. At the furthest point of expansion, the price of housing is exactly equal to the opportunity cost of resources used in production. The price of housing in this jurisdiction, along with the tax and service level, can be used to solve for the constant of integration in Yinger's housing bid model. Yinger (1985) argues that the property tax in this base jurisdiction is distortionary with respect to housing consumption decisions. Variations from this base tax rate, however, are not. These variations will be perfectly capitalized into housing values. Therefore, relatively high property tax rates in one jurisdiction will not repel capital, since the rate will be fully capitalized into immobile factors.

Yinger's conclusions differ from those of the new view in that, in Yinger's model, it is the average tax rate in the metropolitan area that is distortionary. In the new view, it is the average tax rate on capital in the nation that variations are measured against. Mieszkowski and Zodrow (1983) have argued that Yinger's model is in fact consistent with the new view. Unlike the new view models, Yinger does not explicitly model housing production. Agents are choosing a level of housing services that are presumably made up of land and capital; however, he does not explicitly model this. The baseline tax rate distorts the housing decision. Since the capital/land components are not explicitly modeled, there is no prediction as to the effect on the overall return to capital. Yinger does argue that deviations from the baseline tax rate (equivalent to the excise effects of the new view) are completely capitalized. The new view writers are more agnostic about the excise effects. In the new view, these effects generally depend on the model parameters—especially assumptions concerning resident mobility.

Empirical Findings on the Property Tax

This section reviews the empirical literature on capitalization. The approaches taken by other authors, their major findings and some criticisms of the earlier studies are all discussed. This section draws heavily on Bloom, Ladd and Yinger (1983) and Yinger, Bloom, Börsch-Supan and Ladd (1988). These works provide an excellent review of the large number of empirical studies in this area. They also discuss some of the major obstacles associated with estimating the degree of property tax capitalization.

Previous Empirical Work[6]

The authors discussed in this section are primarily interested in discovering the degree to which property tax differentials are capitalized into the price of houses. Recall that theory predicts the that property tax will have two effects. First, the average level of property taxes in a metropolitan area (or a nation, depending on the theory) is expected

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to distort the housing decision. This average level of taxation may change the amount of capital or land used in housing, and in addition, some portion of the tax may be capitalized into the price of houses. The second effect results from differences in the property tax rates on houses within a metropolitan area. These differences can either be interjurisdictional, resulting from different tax rates levied in neighboring communities, or intrajurisdictional, resulting from different tax rates levied on houses within a community as a result of assessment practices.

As discussed, property tax theories are not in agreement as to the degree to which property taxes are capitalized. The new view approach generally models property taxation in a general equilibrium framework similar to the one used by Harberger (1962). The degree of capitalization predicted by the new view depends on assumptions concerning factor mobility, and the elasticities of substitution between capital and land.

An alternative approach is the one followed by Yinger (1982, 1985). Yinger uses an urban model with a central business district. This model predicts that tax differentials between communities will be fully capitalized. The model also predicts the full capitalization of tax differences resulting from assessment errors.

These models generally assume complete information. That is, the models assume that agents have complete knowledge of both the current tax rates and of what the future path of tax rates will be. Of course in practice, complete information is unlikely. In their study of intrajurisdictional capitalization in Massachusetts, Yinger, Bloom, Börsch-Supan and Ladd (1988) find substantially less than full capitalization. They argue that uncertainty surrounding the likely persistence of tax differentials in the communities they study reduces the degree of capitalization.

The studies discussed in this section focus on estimating the capitalization of tax differentials between communities, rather than estimating the effect of the average tax rate on property values. There is generally little or no sample variation in the average tax rate, making it difficult to measure the effect of the average tax rate on housing prices.[7]

Capitalization studies are generally interested in measuring the degree to which property tax differences are capitalized. A property tax differential is said to be fully capitalized, if the difference in price between two otherwise identical properties is equal to the present discounted value (PDV) of the tax differential. The price differential between two properties can usually be observed. Measurements of the PDV of this differential, however, cannot. The PDV is calculated from the price differential using two parameters, the discount rate and the time horizon. The discount rate and the time horizon must be chosen by the researcher and the choices have varied greatly.[8]

Assumptions concerning the discount rate and time horizon can have a large effect on the interpretation of the capitalization results. For example, King (1977) assumes a discount rate of 5% and a time horizon of forty years and states that 67% of property

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tax differentials are capitalized. Yinger, Bloom, Börsch-Supan and Ladd (1988) recalculate the degree of capitalization in King's study using their preferred parameters of a 3% discount rate and an infinite housing life and find the degree of capitalization to be just 36%. Therefore, any estimates of the degree of capitalization will be sensitive to the parameter choices.

Yinger, Bloom, Börsch-Supan and Ladd (1988) recalculate the capitalization findings for many of the studies discussed below assuming a 3% real discount rate and an infinite housing life. This convention is followed for the remainder of this section so that the results of the different authors can be easily compared. The assumption of a 3% discount rate and infinite housing life are rather conservative. Under the assumption of a 3% discount rate and an infinite housing life, 100% capitalization implies that a $1.00 tax differential between houses results in a $33.33 price differential. Likewise under these assumptions, 50% capitalization would imply a $1.00 tax differential leading to a $16.67 price differential.

The capitalization studies can be divided into three broad categories. First, there are "aggregate studies" that use aggregated house price and tax figures such as the median house price and tax rate for a community. This aggregate figure is often used in cases where more detailed housing information is not available. Second, there are "micro" studies that use individual houses as observations. These two types of studies reflect a tradeoff in the types of data that are available. The aggregate studies generally contain a large number of communities, so that there is a large amount of sample variation in the tax rate. However, these studies use an aggregated house value measure that is of lower quality for estimating purposes than individual house observations. Micro studies tend to have a higher quality dependent variable—the actual sales price of individual houses. These studies, however, tend to involve fewer communities so there is less variation in the tax rate. Many micro studies look at only one community and focus on intrajurisdictional capitalization.

Finally, there are studies that take advantage of large scale policy changes, which often serve as "natural" capitalization experiments. These studies have been performed using both aggregated data and individual house observations.

Aggregate Studies

Perhaps the seminal capitalization study is Oates (1969). Oates uses 1960 census data to study fifty-three municipalities in northeastern New Jersey. Oates regresses the median value of owner-occupied houses in each community on the effective property tax rate, the annual expenditure per student in the public schools and a number of additional control variables including: the median number of rooms per house, a proxy for the age of the housing stock, the distance of the community's center from Manhattan and the number of poor families in the community. He finds that a higher tax rate depresses house prices and that increased school spending increases the price of housing. Using a discount rate of 5% and a 40-year time horizon, Oates finds that tax differentials are fully capitalized. Under a 3% discount rate and an infinite house life, the capitalization percentage is 61%.

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Oates recognizes that the tax rate term in his regression may be endogenous. Capitalization theory assumes that higher tax rates lead to lower housing prices. However, a community with relatively low housing values needs a relatively high tax rate to fund a given service level. Therefore, it is difficult to determine the direction of causation between tax rates and housing values. This problem is fundamental to this literature. For a study to credibly state that it has measured the effect of higher taxes on property values, it must first convincingly show that it is measuring capitalization and not the fact that property poor communities need high tax rates to fund services. The same problem is present in reverse when measuring the effects of public expenditures.

To correct for the endogeneity of the tax and expenditure terms, Oates uses two-stage least squares (2SLS). To correctly perform 2SLS, Oates needs variables that are correlated with the tax and expenditure terms, but do not in part determine house prices. Some of the instruments Oates uses include: the median years of school completed by adult males, the population density, the percentage change in population between 1950 and 1960, the percentage of the population enrolled in K-12 schooling and the value of commercial property per resident. It is possible that the value of all of these instruments has some impact on the price of housing in a neighborhood. Therefore, they might not be suitable instruments. In fact, the coefficient on the tax term in Oates's 2SLS regression is the same as the coefficient on the tax term in the ordinary least squares regression, suggesting that if there is an endogeneity problem, the 2SLS estimation does not completely fix it.

Several subsequent authors attempt to improve on Oates's work. King (1977) suggests that Oates mis-specifies the tax term in his regression. The dependent variable in Oates's study is the house price and the tax term used is the tax rate. King argues that if the house price is the dependent variable then the tax *payment* should be used as the independent variable; because for a given tax *rate* increase, the dollar effect will be larger on a higher priced house. Therefore, Oates's specification understates the capitalization for high value houses and overstates it for low value houses. King finds that correcting this specification error reduces the capitalization found with the Oates sample. With a 3% discount rate and infinite house life, Kings capitalization estimate is 36%, an estimate 41% lower than Oates's finding.

Rosen and Fullerton argue that per-student spending is not a good control for service quality. They also use Oates's sample to estimate capitalization after replacing per-student spending with fourth grade student test scores. They find the capitalization rate to be slightly lower than Oates (58% with a 3% discount rate and infinite house life). However, their study contains the tax rate specification error identified by King.

Several authors have attempted to use simultaneous equation systems when estimating the degree of capitalization. Gronberg (1979) uses a six-equation model, with separate equations included for the tax rate and public service expenditure. Using a sample consisting of census data on eighty-three Chicago suburbs in 1970, Gronberg finds that property tax differentials are not capitalized into house values. Dusansky, Ingber and Karatjas (1981) also use a simultaneous equation model. They examine sixty-two

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communities in Long Island, NY using census data. This study also attempts to model and measure the interaction between the rental price of apartments and housing prices. Their estimate of the capitalization rate, assuming a 3% real discount rate and infinite house life, is 22%.

An interesting variation on these studies is Carroll and Yinger (1994). They investigate the effect of property taxes on rents using 1980 Census data for the Boston SMSA. They find that the net burden on tenants from a $1.00 increase in property taxes is only $0.16 on average. Based on this evidence, they conclude that the property tax, at least for renters, cannot be seen as a benefit tax. Some tax shifting does occur because tenants are willing to pay higher rents to receive better services. However, Carrroll and Yinger find that due to the mobility of tenants, the vast majority of the property tax falls on landlords.

The studies using aggregated data all tend to have similar shortcomings. They often use some form of the median or mean house price as the dependent variable. It is not clear how well these average price figures predict the experience of individual houses. These studies generally rely on census data, where the house price reflects the house owner's guess as to what the property is worth. It is possible that owner house price predictions are not very accurate. These studies also run a large risk of omitted variable bias. The median house price in a community is a function of many variables, yet these studies often use only a handful of controls. Finally, these studies generally do not convincingly handle the endogeneity problem. It is difficult to find variables that are potentially correlated with the tax rate but not the price of housing. Therefore, with many of these studies, the possibility that the estimates of tax capitalization are inconsistent remains.

Micro Estimates

Krantz, Weaver and Alter (1982) look at 243 single-family owner-occupied homes that were sold in six Pennsylvania cities and their surrounding suburbs in 1979. The data were gathered from MLS records. They find the capitalization rate to be 20% (using a 3% real discount rate and infinite housing life).

One important aspect of this study is that, unlike the other studies discussed, Krantz, Weaver and Alter do not attempt to correct for the simultaneity of the tax variable. They argue that property taxes will adjust slowly to changes in a community's property values. Therefore, property taxes and house values are not simultaneously determined and ordinary least squares can be used to estimate capitalization. However, the authors provide no evidence to suggest that they are looking at a long run equilibrium, where tax rates have fully adjusted to property values. Therefore, the results of this study should be viewed with some skepticism.

Lea (1982) uses a simultaneous equation model to estimate capitalization. Lea looks at 680 households from the Panel Study of Income Dynamics. Observations from the 1968 survey are merged with local tax and expenditure data for cities and counties

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in which the families are located. Lea finds the capitalization rate to be 26%. Unfortunately, for survey confidentiality reasons, Lea was unable to determine the actual municipality in which his households reside. Instead he only knows the county in which the houses are located. Therefore, he uses average tax and expenditure data for the county in which the household resides. Lea's study may be more appropriate for estimating the effects of the average tax rate in an area than for estimating the effects of tax rate differentials.

Richardson and Thalheimer (1981) examine 861 house sales in Fayette County, Kentucky in 1973 and 1974. These house sales occur in two bordering municipalities that share a school district but have different property tax rates. Richardson and Thalheimer argue that there are few differences in other municipal services between the communities. They find the capitalization rate to be 15% (assuming a 3% real discount rate and infinite house life).

Palmon and Smith (1998a, b) use an innovative methodology to estimate the degree of capitalization. They hypothesize that the ratio of rent to house prices is equal to the sum of the net of user cost of housing and the property tax cost of housing. They use MLS data from an area in Houston where the service level is relatively constant but the tax cost varies widely. They find full capitalization of property tax differentials.

Palmon and Smith's study is compelling both due to their methodological innovation and the quality of their data. However, their results rely on several strong assumptions. First, they use observations of the rental prices of rental housing to estimate the rental price of owner-occupied housing. This estimate is then used to calculate the rent price ratio.

Second, they assume that they can safely omit several variables important to calculating the net user cost of housing, such as the real discount rate, the expected rate of market appreciation and the house's depreciation rate. They argue that these terms will be correlated with observable variables such as the house's age and size that are included in the regression. While it is true that there may be a correlation between the omitted variables and the included variables, it is possible that their regression results are subject to omitted variable bias. Palmon and Smith also estimate the capitalization rate using a more traditional specification and find the rate of capitalization to be 55%.

The micro studies are free from some of the problems of the aggregate studies. The observations are individual houses; therefore, the estimates are free from potential aggregation problems. These studies also tend to have more control variables for both housing and neighborhood characteristics. Therefore, omitted variable bias is less of a concern. Some of these studies use actual sales prices rather than owner estimates of the value, eliminating one more potential source of error. These studies, however, still must contend with the endogeneity issue and face many of the same challenges here as the aggregate studies.

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Natural Experiments

As noted, the most difficult problem encountered by researchers studying capitalization is the potential simultaneity between tax rates and house values. Several authors have exploited particular policy changes to avoid this problem. Statewide property tax reforms can be treated as exogenous to the local communities. Therefore, by examining the change in property values that result from such a tax change, the degree of capitalization can be measured without the usual endogeneity problems.

Rosen (1982) examines the effects of California's Proposition 13 on house prices. Proposition 13 was approved by California voters in 1978. The proposition stated that the annual tax rate on real property could not exceed 1% of the cash value of the property. The cash value of the property was defined as the county assessor's evaluation of the property's value as stated on the 1975–76 tax bill. For property sold after this year, the cash value was defined as the market price. Assessment increases were capped at 2% per year, with properties reassessed to market value on sale. Rosen argues that because of initial state bailouts supporting local public expenditures, Proposition 13 had little effect on service levels in the short run.

Rosen regresses the change in a community's average house sales price on the change in a communities average tax payment and a number of other regressors. His sample consists of sixty-four communities in the San Francisco Bay Area in the years 1978–79. He finds the capitalization rate to be 22% (with a 3% real discount rate and infinite housing life).

Gabriel (1981) examines the effect of Proposition 13 on the average sales price of houses in the San Francisco Bay Area over the same time period as Rosen. He finds the capitalization rate to be somewhat higher than Rosen (36%). Given that Gabriel and Rosen examine the same geographic area over the same period, the large difference in their results is troubling. The divergence of the estimates is most likely due to a specification difference between the two studies. Gabriel suppresses the constant term in his regression, while Rosen does not. The regressions in both studies have the difference in average sales price of houses as the dependent variable, and the difference in annual tax payments as an independent variable. Including a constant term in the regression would capture any differences in the overall price level for housing in the estimating region. Because he suppresses the constant term, Gabriel's tax term is probably capturing some of the effects of the change in the average tax rate.[9] Therefore, the two studies are not measuring the same thing.

A similar large scale tax change is exploited by Yinger, Bloom, Börsch-Supan and Ladd (1988). They examine the effects of court-ordered property revaluation in Massachusetts in the early 1970s. They examine individual home sales before and after revaluation. Their study focuses on the degree of intrajurisdictional property tax capitalization. Their preferred estimate is 21% in one community and 15.8% in another. They conclude that the degree of intrajurisdictional capitalization is likely to vary by community. However, their model predicts that intrajurisdictional capitalization should be complete. They argue that their estimates fall sharply below

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this level because of uncertainty surrounding the reforms. They note, however, that they do not have evidence to support this assertion.

Guilfoyle (1998) uses Michigan's 1994 school finance reforms to examine the effects of interjurisdictional property tax differences on houses values. Michigan's reforms dramatically changed the property tax rate of each community in 1994. Because the property tax change was exogenous to the local communities, there is no problem in determining the direction of causation. Guilfoyle uses a set houses that sold once in 1992 and then again in 1996 to measure the effect of the property tax change. Guilfoyle's estimates of the effects of a $1.00 property tax difference range from $5.00 to $10.00. Under the assumption of a 3% discount rate and an infinite house life, this represents capitalization in the range of 13% to 30%.

Conclusion

The theoretical property tax literature does not offer a clear answer on the effects of the property tax on house prices. The results in the theoretical models are largely driven by assumptions concerning resident mobility and the elasticity of substitution between the various inputs to housing. Therefore, this effect must be measured empirically. Unfortunately, the empirical literature has not yet reached a consensus. Early empirical studies of capitalization generated a wide range of estimates and seemed to suffer from an array of methodological difficulties. More recent studies attempt to correct these difficulties but are still yielding contradictory results. The studies that use natural experiments resulting from statewide property tax reforms (Gabriel, 1981; Rosen, 1982; and Guilfoyle, 1998) find the capitalization effects to be relatively small. The recent study by Palmon and Smith (1998a, b) finds a much higher level of capitalization.

It is unclear if the widely varying results still occurring in the capitalization literature are the result of remaining methodological difficulties or if some other factor is at work. It is certainly possible that the effects of property taxes on house prices is driven in part by some other variable such as the amount of land available for development in a metropolitan area. Areas with little available land may have a more inelastic housing supply curve and therefore experience more tax capitalization. Similarly, there may be factors on the demand side that affect the rate of capitalization. Future capitalization studies should explore whether any other factors are influencing the effect of property taxes on house prices. For now the picture remains unclear.

Endnotes

1. Other assumptions in Tiebout's model include: consumers are fully mobile, consumers have perfect knowledge concerning local revenues and expenditures, a large number of communities exist for consumers to choose among, there are no community-based employment restrictions, there are no externalities associated with local public service provision and communities are sized so that they can produce services at the minimum of their average cost curve.
2. The description of Hobson's results are taken from Mieszkowski and Zodrow (1989).

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3. Note that this formula implicitly assumes an infinite lifetime for housing. Given the long expected lifetime of housing, Yinger argues that this should be a close approximation to the actual present discounted value.
4. Yinger (1982) derives Equation (3) in part by solving a differential equation. The v term represents the constant of integration.
5. Yinger notes that some models of capitalization use flexible boundaries for communities, leading to communities with the most favorable tax and service packages expanding into the territories of less favorable communities. However, he argues that in practice community annexation is relatively rare and community boundaries seldom change.
6. Note that many of the problems with the studies noted here have also been identified and discussed by Yinger, Bloom, Börsch-Supan and Ladd (1988).
7. For an exception that does focus on the average tax rate see Wassmer (1993).
8. Two studies attempt to estimate the real discount rate (Do and Sirmans, 1994; and Palmon and Smith, 1998a). Do and Sirmans begin by assuming full capitalization and a 25-year time horizon and then estimate the discount rate to be 4%. Their estimate of the discount rate relies on the strong assumption of full capitalization. Palmon and Smith estimate capitalization using a procedure that does not requires an assumption regarding the discount rate. They then use their estimates to calculate a real discount rate of 6%.
9. It does not appear that Gabriel is deliberately trying to measure the effects of the average tax rate change. In addition, the tax term in Gabriel's regressions may also be capturing any secular increase in the demand for housing that occurred over the sample period.

References

Aaron, H. J., *Who Pays the Property Tax?* Washington D.C.: Brookings Institution, 1975.

Bloom, H. S., H. F. Ladd and J. Yinger, Are Property Taxes Capitalized Into House Values? In *Local Provision of Public Services: The Tiebout Model After Twenty Five Years*, G. R. Zodrow, (Ed.), New York, NY: Academic Press, Inc, 1983.

Carroll, R. J. and J. Yinger, Is the Property Tax a Benefit Tax? The Case of Rental Housing, *National Tax Journal*, 1994, 47:2, 295–316.

Do, Q. A. and C. F. Sirmans, Residential Property Tax Capitalization: Discount Rate Evidence from California, *National Tax Journal*, 1994, 47:2, 341–48.

Dusansky, R., M. Ingber and N. Karatjas, The Impact of Property Taxation on Housing Values and Rents, *Journal of Urban Economics*, 1981.

Gronberg, T. J., The Interaction of Markets in Housing and Local Public Goods: A Simultaneous Equations Approach, *Southern Economic Journal*, 1979, 46, 445–59.

Gabriel, S. A., Interjurisdictional Capitalization Effects of Proposition 13 in the San Francisco Bay Area, *National Tax Association Proceedings*, 1981, 263–71.

Gravelle, J. G., *The Economic Effects of Taxing Capital Income*, Cambridge, MA: MIT Press, 1994.

Guilfoyle, J. P., The Incidence and Housing Market Effects of Michigan's 1994 School Finance Reforms, Ph.D. Dissertation, Michigan State University, 1998.

Hamilton, B. W., Capitalization of Intrajurisdictional Differences in Local Tax Prices, *American Economic Review*, 1976, 66:5: 743–53.

——., Is the Property Tax a Benefit Tax? In *Local Provision of Public Services: The Tiebout Model after Twenty Five Years*, G. R. Zodrow, (Ed.), New York, NY: Academic Press, Inc., 1983.

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Harberger, Arnold C., The Incidence of the Corporate Income Tax, *Journal of Political Economy*, 70:3, 215–40.

Hobson, P., Land Rents, Optimal Taxation and Local Fiscal Independence in an Economy with Local Public Goods, *Journal of Public Economics*, 1986, 15:1, 59–85.

King, A. T., Estimating Property Tax Capitalization: A Critical Comment, *Journal of Political Economy*, 1977, 85:2, 425–31.

Krantz, D. P., R. D. Weaver and T. R. Alter, Residential Property Tax Capitalization: Consistent Estimates Using Micro-Level Data, *Land Economics*, 1982, 58:4, 488–96.

Lea, M. J., Local Tax and Expenditure Capitalization: Integrating Evidence from the Market and Political Processes, *Public Finance Quarterly*, 1982, 10:1, 95–117.

Mieszkowski, P., The Property Tax: An Excise Tax or a Profits Tax? *Journal of Public Economics*, 1972, 1:1, 73–96.

Mieszkowski, P. and G. R. Zodrow, Taxation and the Tiebout Model: The Differential Effects of Head Taxes, Taxes on Land Rents and Property Taxes, *Journal of Economic Literature*, 1989, 27: 1098–1146.

Netzer, D., *Economics of the Property Tax*, Washington D.C.: Brookings Institution, 1966.

Oates, W. E., The Effects of Property Taxes and Local Public Spending on Property Values: An Empirical Study of Tax Capitalization and the Tiebout Hypothesis, *Journal of Political Economy*, 1969, 77:6, 957–71.

Palmon, O. and B. A. Smith, A New Approach for Identifying the Parameters of a Tax Capitalization Model, *Journal of Urban Economics*, 1998a, 44, 299–316.

——., New Evidence on Property Tax Capitalization, *Journal of Political Economy*, 1998b, 106: 5, 1099–1111.

Richardson, D. H. and R. Thalheimer, Measuring the Extent of Property Tax Capitalization for Single Family Residences, *Southern Economic Journal*, 1981, 48, 674–89.

Rosen, K., The Impact of Proposition 13 on House Prices in Northern California: A Test of the Interjurisdictional Capitalization Hypothesis, *Journal of Political Economy*, 1982, 90:1, 191–200.

Rosen, H. S. and D. J. Fullerton, A Note on Local Tax Rates, Public Benefit Levels and Property Values, *Journal of Political Economy*, 1977, 85:2, 433–40.

Samuelson, P. A., The Pure Theory of Public Expenditures, *Review of Economics and Statistics*, 1954, 36:4, 387–89.

Simon, H. A., The Incidence of a Tax on Urban Real Property, *Quarterly Journal of Economics*, 1943, 59:3, 398–420.

Thomson, P., The Property Tax and the Rate of Interest, In *The American Property Tax*, G. S. Benson et al. (Eds.), Claremont, CA: Lincoln School of Public Finance, 1965.

Tiebout, C. M., A Pure Theory of Local Expenditures, *Journal of Political Economy*, 1956, 64: 5, 416–24.

Wassmer, R. W., Property Taxation, Property Base and Property Value: An Empirical Test of the 'New View,' *National Tax Journal*, 1993, 46:2, 135–59.

Yinger, J., Capitalization and the Theory of Local Public Finance, *Journal of Political Economy*, 1982, 90:5, 917–43.

——., Inefficiency and the Median Voter: Property Taxes, Capitalization, Heterogeneity and the Theory of Second Best, In *Perspectives on Local Public Finance and Public Policy*, Vol. 2, J. M. Quigley (Ed.), Greenwich, CT: JAI Press, 1985.

Yinger, J., H. S. Bloom, A. Börsch-Supan and H. F. Ladd, *Property Taxes and House Values: The Theory and Estimation of Intrajurisdictional Property Tax Capitalization*, San Diego, CA: Academic Press, Inc, 1988.

Zodrow, G. R. and P. Mieszkowski, The Incidence of the Property Tax, In Local Provision of Public Services: The Tiebout Model after Twenty Five Years, George R. Zodrow, (Ed.), New York, NY: Academic Press, Inc 1983.

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